

ARCHITECTURE & MIGRATION DOCUMENT

Global Superstore Executive Dashboard

Technical Replication Specifications for Looker & Tableau

1. Data Model & Architecture

To successfully replicate this Power BI dashboard in Looker (via LookML) or Tableau, the underlying dataset must be modeled as a **Star Schema**. The relationships ensure cross-filtering works accurately without duplicating financial metrics.

TABLE NAME	TYPE	PRIMARY KEY / FOREIGN KEY	DESCRIPTION
Orders	Fact Table	ORDER_ID, REGION	The central transactional table containing granular line-item sales data, revenue, profit, shipping details, and product dimensions.
People	Dimension Table	REGION (Joins to Orders.REGION)	Look-up table mapping sales managers (PERSON) to their respective operating territories.
Returns	Fact/Event Table	ORDER_ID (Joins to Orders.ORDER_ID)	Contains a list of orders that were returned. Used for semi-join counting to calculate return rates against total sales.

2. Global KPI Metrics & Formulas

These are the calculated metrics (Measures) required for the dashboard. Below are their logical definitions for recreation in Tableau Calculated Fields or Looker type: measure blocks.

METRIC NAME	FORMAT	LOGICAL DEFINITION (SQL / DAX EQUIVALENT)
Total Sales	Currency (\$M, \$K)	SUM(Orders[SALES])
Total Profit	Currency (\$K)	SUM(Orders[PROFIT])
Profit Margin %	Percentage (2 Dec)	SUM(Orders[PROFIT]) / SUM(Orders[SALES])

METRIC NAME	FORMAT	LOGICAL DEFINITION (SQL / DAX EQUIVALENT)
Total Orders	Integer (K)	<code>COUNT DISTINCT (Orders[ORDER_ID])</code>
Total Customers	Integer (K)	<code>COUNT DISTINCT (Orders[CUSTOMER_ID])</code>
Return Rate %	Percentage (2 Dec)	<code>COUNT DISTINCT (Returns[ORDER_ID]) / COUNT DISTINCT (Orders[ORDER_ID])</code> <i>*Requires filtering to Orders present in Returns table</i>

3. Page 1: Summary Dashboard

The Summary page provides high-level executive insights, plotting macro-trends across time, geography, and broad product categories. All visuals interact via cross-filtering.

1. Executive KPI Banner

Chart Type: Multi-row Card / Big Number KPIs

Metrics Included: **Total Sales** , **Total Profit** , **Profit Margin %** , **Total Orders** , **Total Customers** , **Return Rate %**

Formatting Note: Metrics are displayed horizontally across the top with descriptive subtitles.

2. Total Sales and Total Profit by Year and Quarter

Chart Type: Dual-Axis Line Chart

X-Axis (Time): **ORDER_DATE** (Year, Quarter)

Y-Axis (Left): **Total Sales** (Light Blue Line)

Y-Axis (Right): **Total Profit** (Dark Blue Line)

Interaction: Clicking a data point filters the rest of the page to that specific quarter.

3. Total Sales by MARKET

Chart Type: Horizontal Bar Chart

Y-Axis (Rows): MARKET (e.g., APAC, EU, US)

X-Axis (Length): Total Sales

Sort: Descending by Total Sales

4. Total Sales by CATEGORY and SUB_CATEGORY

Chart Type: Treemap

Grouping Level 1: CATEGORY (Technology, Furniture, Office Supplies)

Grouping Level 2: SUB_CATEGORY (Phones, Chairs, Binders, etc.)

Size/Value: Total Sales

5. Total Sales by SEGMENT

Chart Type: Donut Chart

Slices (Color): SEGMENT (Consumer, Corporate, Home Office)

Angle/Value: Total Sales

Labels: Category + Data Value + Percentage (e.g., \$2.50M (51.34%))

4. Page 2: Sales Operations

This page provides detailed operational insights for managers. It features advanced DAX filtering (Top/Bottom products) and strategic scatter plotting for risk analysis.

Page-Level Filters

- **On-Page Slicers:** REGION (Dropdown), SEGMENT (Dropdown), ORDER_DATE (Range Slider).

1. Extreme Performers (Top 5 & Bottom 5 Products)

Chart Type: Clustered Bar Chart

Y-Axis: PRODUCE_NAME

X-Axis: Total Sales

Advanced Configuration (The "Missing Middle"): This chart hides average products. In Tableau, this requires two sets (Top 5 Set, Bottom 5 Set) combined. In Looker, requires a custom derived table ranking.

```
// Power BI DAX equivalent used:  
Top 5 and Bottom 5 Filter =  
IF(ISBLANK([Total Sales]) || [Total Sales] = 0, 0,  
    VAR RankTop = RANKX(FILTER(ALLSELECTED(Orders[PRODUCE_NAME]),  
        [Total Sales] > 0), [Total Sales], , DESC)  
    VAR RankBottom = RANKX(FILTER(ALLSELECTED(Orders[PRODUCE_NAME]),  
        [Total Sales] > 0), [Total Sales], , ASC)  
    RETURN IF(RankTop <= 5 || RankBottom <= 5, 1, 0)  
)
```

Color Rule: Top 5 = Dark Blue. Bottom 5 = Warning Red.

2. Sub-Category Strategic Quadrant

Chart Type: Scatter Plot

X-Axis: **Return Rate %** (Risk metric)

Y-Axis: **Profit Margin %** (Health metric)

Detail/Bubble: **SUB_CATEGORY**

Bubble Size: **Total Sales**

Analytics Lines: X-Axis Constant Line at average Return Rate. Y-Axis Constant Line at average Profit Margin. (Creates 4 strategic quadrants).

3. Manager Operations Scorecard

Chart Type: Data Matrix / Pivot Table

Rows: **REGION** (Hierarchy expands to `People.PERSON`)

Columns/Values: **Total Sales** , **Total Orders** , **Return Rate %**

Conditional Formatting:

- Total Sales features light-blue background Data Bars.
- Return Rate % features font-color warnings (Red for high rates).

4. Operational Flow Charts (Bottom Center/Right)

Chart A: Total Sales by Year and SEGMENT

- Type: Stacked Column Chart
- X-Axis: ORDER_DATE (Year) | Y-Axis: Total Sales | Legend/Stack: SEGMENT

Chart B: Total Orders by SHIP_MODE

- Type: Donut Chart
- Color: SHIP_MODE | Value: Total Orders